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No.: Tugas 8 TF-48-03 Date:

1.) Dik:

$t: 0, 1, 2, 3, 4, 5, 6$

$F(t): 0.0, 3.2, 5.1, 6.0, 5.4, 3.1, 0.4$ kN

$h: 1$ $m: 50$ kg

a.) Metode trapesium

$$J_T : \frac{h}{2}[F_0 + 2F_1 + 2F_2 + 2F_3 + 2F_4 + 2F_5 + F_6]$$

$$: \frac{1}{2}[0.0 + 2(3.2) + 2(5.1) + 2(6.0) + 2(5.4) + 2(3.1) + 0.4]$$

$$: \frac{1}{2}[0.0 + 6.4 + 10.2 + 12.0 + 10.8 + 6.2 + 0.4]$$

$$: \frac{1}{\cancel{2}}(\cancel{46}.0)$$

$$: 23.0 \text{ kNs}$$

b.) Metode simpson 1/3

$$J_S : \frac{h}{3}[F_0 + 4F_1 + 2F_2 4F_3 + 2F_4 + 4F_5 + F_6]$$

$$: \frac{1}{3}[0.0 + 4(3.2) + 2(5.1) + 4(6.0) + 2(5.4) + 4(3.1) + 0.4]$$

$$: \frac{1}{3}[0.0 + 12.8 + 10.2 + 24.0 + 10.8 + 12.4 + 0.4]$$

$$: \frac{1}{3}(70.6)$$

$$: 23.5333 \text{ kNs}$$

1kN : 1000 Ns

J_T : 23.0 x 1000 = 23000 Ns

J_S : 25.5333 x 1000 = 23533.3333 Ns

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$\Delta v = \frac{J}{m}$

$\Delta v_T = \frac{23000}{50}$ $\Delta v_S = \frac{23333.3333}{50}$

$\Delta v_T = 460 \, m/s$ $\Delta v_S = 470.6667 \, m/s$

$\text{Selisih relatif} = \left| \frac{J_S - J_T}{J_S} \right| \times 100\%$

$= \left| \frac{23.5333 - 23.0}{23.5333} \right| \times 100\%$

$= \frac{0.5333}{23.5333} \times 100\%$

$= 2.2663\ \%$

2.) Dik:

$h = 0.1 \, mm$

X mm	-0.3	-0.2	-0.1	0.0	0.1	0.2	0.3
U(x) μJ	0.500	0.220	0.055	0.000	0.060	0.250	0.590

a.) Selisih pusat untuk $\frac{dU}{dx}$ di $x = 0.0$

$$\left. \frac{dU}{dx} \right|_{x=0} = \frac{U(0.1) - U(-0.1)}{2h}$$

$$= \frac{0.060 - 0.055}{2(0.1)}$$

$$= \frac{0.005}{0.2} = 0.025 \, \mu J/mm$$

b.) Gaya F (0.0)

$$F(x) = -\frac{dU}{dx}$$

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$1\,MJ/mm = 1\,mN$

$F(0.0) = -0.025\,mN$

c.) Selisih pusat untuk $\frac{d^2U}{dx^2}$ di $x = 0.0$

$$\left.\frac{d^2U}{dx^2}\right|_{x=0} = \frac{U(0.1) - 2U(0.0) + U(-0.1)}{h^2}$$

$$= \frac{0.060 - 2(0.000) + 0.055}{(0.1)^2}$$

$$= \frac{0.115}{0.01} = 11.5\ MJ/mm^2$$

$$= 11.5\ N/m$$

d.) Selisih maju di $x = -0.3$

$$\left.\frac{dU}{dx}\right|_{x=-0.3} = \frac{U(-0.2) - U(-0.3)}{h}$$

$$= \frac{0.220 - 0.500}{0.1}$$

$$= \frac{-0.280}{0.1} = -2.8\ MJ/mm$$

turunan kedua:

$$\left.\frac{d^2U}{dx^2}\right|_{x=-0.3} = \frac{U(-0.1) - 2U(-0.2) + U(-0.3)}{h^2}$$

$$= \frac{0.055 - 2(0.220) + 0.500}{(0.1)^2}$$

$$= \frac{0.055 - 0.440 + 0.500}{0.01}$$

$$= \frac{0.115}{0.01} = 11.5\ MJ/mm^2$$

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e.) Selisih mundur di $x: 0,3$

$$\left.\frac{dU}{dx}\right|_{x:0.3} : \frac{U(0.3) - U(0.2)}{h}$$

$$: \frac{0.590 - 0.250}{0.1}$$

$$: \frac{0.340}{0.1} : 3.4 \ \mu J/mm$$

turunan kedua :

$$\left.\frac{d^2U}{dx^2}\right|_{x:0,3} : \frac{U(0.3) - 2U(0.2) + U(0.1)}{h^2}$$

$$: \frac{0.590 - 2(0.250) + 0.060}{(0.1)^2}$$

$$: \frac{0.590 - 0.500 + 0.060}{0.01}$$

$$: \frac{0.150}{0.01} : 15.0 \ \mu J/mm^2$$

F.) Di titik $x = 0.0$:

$$\frac{d^2U}{dx^2} = 11.5 > 0$$

karena turunan kedua bernilai positif, titik $x = 0.0$ berada di sekitar minimum energi potensial. Jadi, titik tersebut dapat dianggap sebagai titik stabil.

3.) Dik:

$$\frac{dv}{dt} : 9.81 - 0.08v^2$$

$$F(t.v): 9.81 - 0.08v^2$$

$$v(0): 0$$

$$h: 0.5s$$

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Titik waktu:

$t_0 = 0 \quad t_1 = 0.5 \quad t_2 = 1.0 \quad t_3 = 1.5 \quad t_4 = 2.0$

a.) Metode Euler

$$V_{n+1} = V_n + hF(t_n, V_n)$$

$$V_{n+1} = V_n + 0.5(9.81 - 0.08V_n^2)$$

n	t_n s	V_n m/s	$F(t_n, V_n)$ m/s²	V_{n+1} m/s
0	0.0	0.0000	9.8100	4.9050
1	0.5	4.9050	7.8853	8.8476
2	1.0	8.8476	3.5475	10.6214
3		10.6214	0.7849	11.0138

$V_1 = 0 + 0.5(9.81) = 4.9050$

$V_2 = 4.9050 + 0.5(7.8853) = 8.8476$

$V_3 = 8.8476 + 0.5(3.5475) = 10.6214$

$V_4 = 10.6214 + 0.5(0.7849) = 11.0138$

$V(2.0)$ Euler $= 11.0138$ m/s

b.) Metode Runge - Kutta Orde 4

$k_1 = F(t_n, V_n)$

$k_2 = F\left(t_n + \frac{h}{2}, V_n + \frac{h}{2}k_1\right)$

$k_3 = F\left(t_n + \frac{h}{2}, V_n + \frac{h}{2}k_2\right)$

$k_4 = F(t_n + h, V_n + hk_3)$

$$V_{n+1} = V_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

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$\frac{h}{2} = 0.25 \quad \frac{h}{6} = \frac{0.5}{6} = 0.0833333$

$k_1 = F(0.0) = 9.81 - 0.08(0)^2 = 9.8100$

$k_2 = F(0.25, 0 + 0.25(9.81))$

$k_2 = F(0.25, 2.4525)$

$k_2 = 9.81 - 0.08(2.4525)^2 = 9.3288$

$k_3 = F(0.25, 0 + 0.25(9.3288))$

$= F(0.25, 2.3322)$

$k_3 = 9.81 - 0.08(2.3322)^2 = 9.3749$

$k_4 = F(0.5, 0 + 0.5(9.3749))$

$= F(0.5, 4.6874)$

$k_4 = 9.81 - 0.08(4.6874)^2 = 8.0522$

$V_1 = 0 + \frac{0.5}{6}(9.8100 + 2(9.3288) + 2(9.3749) + 8.0522)$

$= 4.6058 \text{ m/s}$

$V(2.0)_{RK4} = 10.4425 \text{ m/s}$